Hadronic-Coefficient Simplification in FACET

A measured NLO and NNLO comparison of algebraic orderings

Internal calculation record

9 August 2026

Abstract

This report reconstructs the hadronic-coefficient simplification calculations performed during 7–8 August 2026 and records their consolidation into a process-card-driven FACET routine. It distinguishes algebra done on amplitude-pair coefficients, complete unreduced integral coefficients, and final master-integral coefficients. Every quoted size is identified as either Mathematica `ByteCount` or an actual serialized file size. At NLO, simplifying complete target coefficients before inserting the Kira reduction rules is the best measured ordering; a bounded master-level cleanup is then inexpensive. At NNLO, constructing a monolithic master coefficient is not viable. The measured alternative retains additive contributions, performs exact rational algebra in positive momentum-fraction roots, and certifies the common Laurent monomial only after cancellations. NLO and NNLO now use one physical normalization declared by the process card; they differ only in whether a coefficient is supplied already assembled or as additive contributions. The reviewed implementation separately certifies the source kinematic chamber and its dimensionless image, verifies the mass dimension and invertibility of the coordinate map, and freezes branch-sensitive analytic objects while rational coefficient algebra is performed. The report also isolates the TT BMHV dimension-bookkeeping error and separates coefficient algebra from the still-open analytic evaluation of NNLO cut masters.

Contents

1 Scientific object and scope

The object being reduced is an exact coefficient multiplying a cut master integral. For an amplitude labelled by p and its conjugate labelled by q, write

$$\mathcal{A}_p\mathcal{A}_q^* = \sum_T c_T^{(pq)}\, T, \qquad\qquad T = \sum_m R_{Tm}\, M_m. \tag{1}$$

Here T is a cut integral before IBP reduction, M_m is a master integral, and R_{Tm} is the exact Kira reduction coefficient. The final coefficient of M_m is therefore

$$C_m = \sum_{p,q,T} c_T^{(pq)} R_{Tm}. \tag{2}$$

This identity defines three distinct places where algebra can be attempted:

1. simplify each $c_T^{(pq)}$ before summing amplitude pairs;
2. form $c_T = \sum_{p,q} c_T^{(pq)}$, simplify it, and then insert R_{Tm};
3. form the complete C_m and simplify only after IBP substitution.

Let the nontrivial momentum fractions be $\boldsymbol{\xi} = (\xi_1, \ldots, \xi_r)$. After exact hadronic-coordinate substitution, every nonzero coefficient accepted by the simplifier must have the form

$$C_m = \mathcal{D}(\boldsymbol{\xi})\boldsymbol{\xi}^{\boldsymbol{\nu}} H_m(I, \epsilon), \qquad \boldsymbol{\xi}^{\boldsymbol{\nu}} \equiv \prod_{i=1}^{r} \xi_i^{\nu_i}, \tag{3}$$

where $\mathcal{D}$ is the product of twist-two distributions, I denotes the chosen partonic invariants, and H_m contains neither momentum fractions nor forbidden hadronic vectors. The card may state $\mathcal{D}$ and $\boldsymbol{\nu}$ explicitly, or request their exact derivation.

For the ppHX UU example this gives

$$\mathcal{P}_{\mathrm{UU}} = \frac{f_1(x_a)f_1(x_b)D_1(z_h)}{x_a x_b z_h^2}, \qquad C_m = \mathcal{P}_{\mathrm{UU}}\, H_m(s, t, u, \epsilon), \tag{4}$$

up to coupling, color, and conventional powers of π. Equation (4) is an example supplied by the ppHX card, not an assumption embedded in the reusable simplifier.

This report concerns coefficient algebra. It does not claim that the 342 NNLO cut masters have been evaluated analytically, nor that endpoint and distributional reconstruction has been completed.

2 What every quoted size means

Several earlier notes used the word "bytes" for different objects. The distinction is essential:

Mathematica `ByteCount`
An in-kernel estimate of the expression tree. It is meaningful for comparisons in the same Mathematica version.

Serialized bytes
The exact `FileByteCount` of a saved `.wl` or binary artifact. It measures I/O and disk use and depends on the chosen serialization grammar.

Additive terms The number of summands after splitting only the top-level `Plus`. A different outer factorization changes this count.

Objects Independently scheduled pair, target, or master coefficients. Counts at different algebraic stages are not comparable.

Target leaves Individual Kira-rule contributions entering a selected master column.

Fractions Stored exact numerator-denominator pairs after local hadronic cleanup.

No reduction ratio in this report mixes `ByteCount` with serialized bytes.

3 Physical substitution and branch control

3.1 The process-card contract

Every card used for hard-coefficient simplification contains a `CoefficientKinematics` block declaring

1. the positive fractions $\boldsymbol{\xi}$, either explicitly or by exact inference from the card's momentum-fraction map;
2. the expected distribution product $\mathcal{D}$ and Laurent valuation $\boldsymbol{\nu}$, or `Automatic` for exact derivation;
3. the physical invariant chamber;
4. a positive scale Q^2 and an invertible map to dimensionless invariant coordinates;
5. variables forbidden in H_m, with automatic inclusion of all declared fractions and hadronic basis vectors;
6. the allowed branch grammar.

The routine rejects a missing block, a missing declaration, a noninvertible coordinate map, a dimensionful coordinate, a scale whose positivity is not implied by the card, and an explicit fraction list that differs from the momentum-fraction map. It also requires the source chamber and the dimensionless chamber to be separately nonempty. If $X_i = X_i(I_1, \ldots, I_n)$ denotes the forward coordinate map and $I_a = I_a(Q^2, X_1, \ldots, X_n)$ its inverse, the exact checks are

$$X_i(I(Q^2, X)) = X_i, \qquad I_a(Q^2, X(I)) = I_a, \tag{5}$$

$$\mathcal{R}_I \Longrightarrow \mathcal{R}_X(X(I)), \qquad \mathcal{R}_X \Longrightarrow \mathcal{R}_I(I(Q^2, X)), \tag{6}$$

where $\mathcal{R}_I$ and $\mathcal{R}_X$ are the source and dimensionless physical regions. They are never conjoined before the coordinate equations are imposed; this prevents an inconsistent card from making an equality check vacuously true.

For ppHX, the NLO UU and TT cards contain exact coordinates for

$$P_a,\ P_b,\ P_h,\ n_h,\ \bar{n}_h,\ S_T,\ S_{hT} \tag{7}$$

in terms of s, t, u, x_a, x_b, z_h and the spin azimuths. The physical chamber contains

$$s > 0, \qquad t < 0, \qquad u < 0, \qquad s + t + u > 0, \qquad 0 < x_a, x_b, z_h < 1, \tag{8}$$

together with the required reality assumptions. The package infers $0 < \xi < 1$ for each nonempty momentum fraction declared in the card and combines those inequalities with the card assumptions.

3.2 Why known signs do not remove branch bookkeeping

There is no physical ambiguity about the signs in Eq. (8). The algebraic issue is that Mathematica does not globally replace, for example,

$$\sqrt{x_a x_b} \longrightarrow \sqrt{x_a}\sqrt{x_b} \tag{9}$$

unless positivity is built into the transformation. The identity is false on a general complex domain. A global `PowerExpand` would therefore erase information that the final analytic result must retain.

The implemented branch grammar, `PositiveMonomialRoots`, introduces one positive root r_i for each declared fraction,

$$\xi_i = r_i^2, \qquad r_i > 0, \qquad i = 1, \ldots, r. \tag{10}$$

Only a certified Laurent monomial in the declared fractions is lifted. Thus

$$\prod_{i=1}^{r} \xi_i^{n_i/2} \longmapsto \prod_{i=1}^{r} r_i^{n_i}. \tag{11}$$

A nonmonomial fraction base such as $\xi_1 + \xi_2$ is rejected. A logarithm, Gamma function, hypergeometric function, absolute value, step function, or distribution whose argument depends on a fraction is also rejected rather than transformed. Fraction-independent noninteger powers, logarithms, Gamma functions, angular functions, distributions, and BMHV tensors are replaced by inert symbols while the rational coefficient is simplified and are restored afterward. Exact equality is checked before and after this freezing; no `PowerExpand` is used.

For an additive coefficient $C_m = \mathcal{D} \sum_j c_{mj}$, write each lifted contribution as $c_{mj} = N_{mj}(\boldsymbol{r})/D_{mj}(\boldsymbol{r})$ and form the exact sum

$$\sum_j c_{mj} = \frac{N_m(\boldsymbol{r})}{D_m(\boldsymbol{r})}. \tag{12}$$

The Laurent valuation is accepted only if there is an integer vector $\boldsymbol{n}$ and a fraction-independent H_m such that

$$N_m(\boldsymbol{r}) = H_m\, \boldsymbol{r}^{\boldsymbol{n}} D_m(\boldsymbol{r}), \qquad \boldsymbol{\nu} = \frac{1}{2}\boldsymbol{n}. \tag{13}$$

The identity in Eq. (13) is checked coefficient by coefficient in the sparse root polynomial. This is why square-root terms may cancel between contributions without being discarded prematurely.

The scale and coordinates are likewise card data. In the ppHX cards they are

$$x = -\frac{t}{s}, \qquad y = -\frac{u}{s}, \qquad s > 0, \quad x > 0, \quad y > 0, \quad x + y < 1, \tag{14}$$

and the routine derives the power p_m from exact homogeneity:

$$H_m(s, t, u, \epsilon) = s^{p_m} \widehat{H}_m(x, y, \epsilon). \tag{15}$$

No common value of p_m is assumed across masters, and p_m may depend on ϵ. The inverse coordinate map and the reconstruction of the dimensional coefficient are checked exactly.

4 NLO UU: where simplification should occur

4.1 Complete 5 by 5 calculation

The complete 25-pair calculation gives six masters. Table 1 records the measured stages.

The extracted prefactor was

$$\frac{C_F\, \pi^{3+\epsilon} \alpha_s^3 D_1(z_h) f_1(x_a) f_1(x_b)}{x_a x_b z_h^2}. \tag{16}$$

All six remaining coefficients were exactly free of momentum fractions and hadronic basis vectors.

Table 1: NLO UU 5 by 5 calculation. The final size is an actual serialized file size; intermediate input sizes were not retained in this compact summary.

Quantity	Measured result
Pair construction	13.3465 s
Pair store	1,086,418 serialized bytes
Kira reduction	50.1824 s
Unreduced targets	77
Masters	6
Hadronic substitution	2.01 s
Complete-target cleanup	6.65 s
Complete-master cleanup	9.04 s
Complete coefficient stage	48.5528 s
Final result	170,483 serialized bytes

4.2 The four orderings on the 10 by 10 sample

The four measured orderings are summarized in Table 2.

Table 2: NLO UU orderings. All sizes are Mathematica `ByteCount`. The target-then-master time is the second stage only.

Ordering	Objects	Input	Output	Time	Result after composition
Pair first	1332	89,111,000	2,694,944	43.403 s	Master size 1,574,064
Target first	117	46,940,920	1,362,792	23.715 s	Master size 1,548,952
Master first	1 tested	–	–	> 281 s	One coefficient unfinished
Target then master	8	1,646,272	1,036,008	62.125 s	Final size 1,041,912

Target first is the best first expensive operation because pair-to-pair cancellations have already occurred while the Kira map has not yet inflated the expression. Pair first repeats algebra. Master first presents the simplifier with the largest sum. Target then master gives the smallest final object, but the sequential time is

$$23.715\,\mathrm{s} + 62.125\,\mathrm{s} = 85.840\,\mathrm{s}, \tag{17}$$

not 62.125 s as stated in the earlier narrative.

The compact benchmark summary did not finish its independent equality test between all four routes. The timing and size comparison is valid; exactness claims in the final NLO calculation instead come from the later production calculation and the certified fraction-normalization tests.

4.3 Detailed NLO algebraic comparisons

The largest target illustrates why a generic rational normal form is not automatically a compact physics form:

Method	Input `ByteCount`	Output `ByteCount`	Time
Termwise `Simplify`	3,517,376	49,016	4.838 s
Whole `Simplify`	3,517,376	327,568	86.378 s
Whole `FullSimplify`	3,517,376	no result	120.045 s
`FactorTerms[Cancel[Together]]`	3,517,376	6,812,872	50.350 s

The isolated equality tests for the first two entries reached their time bounds, so those two rows establish timing and representation size, not a separate exact identity. The complete later route was checked exactly.

The best NLO factor-first chain was:

$$1{,}362{,}792 \xrightarrow[3.393\,\mathrm{s}]{\text{extract common factor}} 684{,}744, \tag{18}$$

$$855{,}104 \xrightarrow[10.741\,\mathrm{s}]{\text{final master cleanup}} 759{,}840, \tag{19}$$

where all sizes are `ByteCount`. Including the preceding target cleanup gives about 37.76 s. Repeating the master cleanup for 9.426 s left the size unchanged.

Direct target cleanup without factor extraction needed 450.188 s to change 1,542,536 to 727,376 bytes. By contrast, certified division by the universal fraction monomial followed by `Simplify` changed 1,035,984 to 283,144 bytes in 5.877 s. A further `FullSimplify` gave only a modest reduction to 256,456 bytes in 32.352 s. `Factor`, color collection, signature grouping, and denominator grouping all expanded the expression. The complete measurements appear in Appendix A.

5 NLO TT: correction, branch structure, and timing

5.1 The BMHV error

TT contains evanescent tensor monomials. Let a loop-independent factor from the original trace be $F(D)$, and let $\mathcal{T}_r^{(D)}$ denote the rank $2r$ evanescent tensor integral. The correct bookkeeping is schematically

$$F(D)\,\mathcal{T}_r^{(D)} \longrightarrow F(D)\sum_j a_j(D+2r)I_j^{(D+2r)}, \tag{20}$$

followed by dimensional recurrence for the scalar integrals. The incorrect code instead changed the inherited factor to $F(D+2r)$.

The diagnostic rank-two angular identity is

$$\left\langle (k\cdot Y)^2 \right\rangle_N = -\frac{k_\perp^2}{N-3}, \qquad Y^2=-1. \tag{21}$$

At tensor stage zero, $N=D$ and the denominator is $D-3$. After one dimension shift, $N=D+2$ and the denominator is $D-1$. A pre-existing loop-free factor such as $D-4$ remains $D-4$. Exact tests checked rank one, rank two, cut projection, both denominators, preservation of the original factor, and rejection of the incorrect denominator.

The practical consequence was large:

Route	`ByteCount`	GLI occurrences	Distinct GLIs
Incorrect degree-zero tensor route	184,796,536	39,036	31
Correct sparse cut route	27,754,864	9	9

The TT code therefore did require a physics correction before simplification measurements were meaningful.

5.2 TT coefficient algebra

For the 5 by 5 sample, target cleanup changed 139,298,664 to 10,562,384 `ByteCount` bytes in 52.02 s on four kernels. Final-master cleanup took 30.36 s. The final six-master file occupies 750,504 serialized bytes. The angular dependence closes on

$$\{\cos\phi_a\cos\phi_h,\ \sin\phi_a\sin\phi_h\}\,, \tag{22}$$

and both mixed sine-cosine structures vanish exactly.

For four representative TT targets, `Together` was fastest but returned the larger rational form. `Simplify`, `Together` followed by `Simplify`, and the historical collect routine all reached the same smaller size. Reversing the order, `Simplify` followed by `Together`, returned the larger form. For example, the 6,438,104-byte target changed to 22,848 bytes in 1.528 s with `Simplify`, to the same size in 0.836 s with `Together` then `Simplify`, but to 74,048 bytes in 0.480 s with the reverse order.

An explicit overall i in an intermediate TT coefficient is not, by itself, evidence of a wrong square-root branch. It may originate from the amplitude, cut, or projector convention. Reality must be checked only after the full normalization and angular projection. The positive-root rules address the separate issue of algebraically valid radical identities.

6 NNLO: why monolithic master simplification fails

The NNLO double-real UU reduction contains 44,877 Kira targets and 342 masters. The earlier post-IBP master files entering the old final cleanup occupy 3,960,695,102 serialized bytes; the resulting files occupy 3,597,827,125 bytes. This is only a 1.101-fold reduction.

6.1 Target-level tests

The 64 largest target coefficients changed from 83,386,848 to 28,279,424 `ByteCount` bytes in 75.643 s on four kernels. On the largest isolated target, whole `Simplify` gave

$$2{,}434{,}656 \to 811{,}576 \quad (15.30\,\mathrm{s}), \tag{23}$$

while termwise `Simplify` gave

$$2{,}434{,}656 \to 701{,}328 \quad (6.92\,\mathrm{s}). \tag{24}$$

Both isolated differences were exactly zero. Pure rational normalization was faster but expanded the target: `Cancel[Together]` produced 3,748,912 bytes, and adding `FactorTerms` produced 5,316,312 bytes.

A stratified set of 125 complete targets changed from 52,630,888 to 559,984 `ByteCount` bytes in 30.45 s on four kernels. The fraction dependence consisted only of explicit half-integer Laurent powers; no fraction-dependent logarithm, Gamma function, or hypergeometric function was found in that sample.

6.2 Final-master tests

Eight whole-master `Simplify` jobs each reached 1,800 s and 4 GiB without producing retained output. Aggregate memory was approximately 34 GiB. Termwise simplification of master 8 was feasible: 377 terms changed a 4,655,210-byte serialized file to 3,082,976 bytes in 175.070 s on eight kernels.

The rational/signature decomposition gave only moderate serialized reductions:

Master	Input	Output	Time
1	1,607,146,927	920,945,177	about 19.5 min
2	903,734,615	464,281,765	about 15 min
4	83,696,010	47,987,161	68.55 s

Master 4 had 2,591 terms and 2,591 different signatures, so there was no cross-term signature merge. The gain came only from rational normalization inside individual terms.

Symbolica did not improve the tested coefficient: direct Mathematica took 5.91 s and produced 0.70 MB, while Symbolica followed by Mathematica took 12.16 s and produced 0.81 MB before including preprocessing time. No exact end-to-end FORM or SymPy route was established for the full grammar.

7 NNLO: exact entrywise and denominator algebra

7.1 Entrywise positive-root cleanup

Three post-IBP master columns were selected by incoming target-leaf count: one small, one median, and the largest column. Together they contain 38,460 target leaves in 38,400 exact denominator entries. The source occupies 859,107,253 serialized bytes. Applying the root lift (10), proving the universal Laurent factor entry by entry, and reducing in the positive-s root ring produced a 131,787,741-byte store in 1,304.639 s on eight kernels, a 6.52-fold reduction.

The independent audit checked source hashes, numerator and denominator hashes, entry and leaf counts, exact arithmetic, nonzero denominators, absence of momentum fractions and temporary roots, and integer powers of s.

7.2 Equal-denominator assembly

For a class of exactly equal denominators,

$$\sum_i \frac{n_i}{d} = \frac{\sum_i n_i}{d}. \tag{25}$$

A hash only locates candidate classes; structural equality decides whether Eq. (25) is used.

Always retaining the cancelled form was a mistake for storage. For the largest master, the fraction count fell from 38,306 to 6,290 in 6,375.22 s, but the fraction-list `ByteCount` grew from 422,282,520 to 1,727,567,608 and the serialized output occupied 488,886,853 bytes.

The corrected size-monotone rule constructs

$$\text{raw} = \left\{ \sum_i n_i, d \right\}, \tag{26}$$

$$\text{cancelled} = \text{NumDen}\left[\text{Cancel}\left(\frac{\sum_i n_i}{d}\right)\right] \tag{27}$$

and retains the smaller exact pair according to `ByteCount`. The largest master then changed from 38,306 to 6,557 fractions in 1,260.216 s. Its fraction-list size changed from 422,282,520 to 351,559,840 bytes and its serialized output occupied 99,892,985 bytes.

7.3 Dimensionless normalization

The exact homogeneity transformation in Eq. (14) found scale powers $p_m = 4, 2, -2$ for the three selected masters. For the largest master it changed 351,559,840 to 296,802,344 `ByteCount` bytes and 99,892,985 to 92,385,524 serialized bytes in 430.914 s on eight kernels.

A second size-monotone denominator merge changed 6,557 to 5,973 fractions and 296,802,344 to 296,275,768 `ByteCount` bytes in 904.575 s. The final serialized file occupies 92,241,826 bytes. A complete independent replay took 943.22 s.

Factoring every dimensionless denominator over $\mathbb{Q}(\epsilon)$ took 27.94 s and found only 30 distinct x, y-dependent factors. However, a separate factor dictionary occupied 91,385,453 serialized bytes, only 0.93 percent less than the direct fraction list. It is useful for identifying the singular locus, but not as a new storage representation.

8 Is the simplification post-IBP?

The answer depends on the row being discussed:

- Pair-first and target-first simplification act before the Kira rules are inserted into Eq. (2), although the rules have already been computed.
- Master-first, target-then-master's second stage, every final-master comparison from 7 August, and the complete NNLO selected-column root-ring study act after the IBP rules have been inserted.
- The NNLO selected-column entries are not unreduced integrals. They are the individual post-IBP contributions $c_T R_{Tm}$ to a fixed master coefficient C_m.

Thus the measurements used to estimate the final 342 coefficients are post-IBP master-coefficient algebra, as expected.

9 Card-driven implementation and exact tests

The reusable implementation is now contained in `FeynFacet/Private/Simplification.wl`. It contains no literal ppHX fraction names, distribution product, Laurent valuation, invariant ratios, or selected master labels. These enter through the card contract stated above. Historical benchmark drivers still contain ppHX-specific choices because they record the measurements in this report; they are not the maintained simplification interface.

The public routine `SimplifyHardCoefficients` applies the same physical normalization to two exact input forms:

$$\begin{aligned} \text{assembled:} \quad & \{C_1, \ldots, C_N\}, \\ \text{contribution-wise:} \quad & \{\{c_{11}, \ldots, c_{1n_1}\}, \ldots, \{c_{N1}, \ldots, c_{Nn_N}\}\}, \quad C_m = \sum_j c_{mj}. \end{aligned} \tag{28}$$

The second form is needed when constructing C_m first would exceed memory or when the cancellation in Eq. (13) must be exposed without a global simplification. It is not an NNLO-specific definition. Either form may be used at any perturbative order.

The acceptance criteria in the exact tests were: the card schema is complete; the source and dimensionless chambers satisfy Eq. (6); every coordinate is dimensionless; the distribution product and Laurent valuation are derived or agree with their explicit declarations; no forbidden variable remains; every scale-separated coefficient reconstructs the dimensional expression exactly; and every result contains exact data only. A renamed synthetic process used $\{\rho_A, \rho_B, \rho_H\}$, scale Q^2, and coordinates X, Y. It also included two contributions proportional to $\sqrt{\rho_A \rho_B}$ that cancel only after addition. The routine derived

$$\boldsymbol{\nu} = (-1, -1, -2) \tag{29}$$

and reconstructed both hard coefficients exactly. The negative tests reject a missing or incomplete block, contradictory coordinate chamber, dimensionful coordinate, incomplete coordinate map, wrong twist-two channel, reciprocal distribution factor, wrong Laurent valuation, residual parton momentum, nonmonomial fraction root, and fraction-dependent analytic function.

Stored physics expressions gave the following measured results:

Input	Exact coefficients	Time	Result `ByteCount`
NLO UU	6	48.51 s	859.88 KiB
NLO TT, smallest retained coefficient	1	2.58 s	67.16 KiB
NNLO UU, one stored target	1 (2 contributions)	2.34 s	27.91 KiB

For all three, the extracted distribution factor agreed with the card value. The NLO cards declare and verify

$$\mathcal{D}_{\text{extracted}} = \mathcal{D}_{\text{card}} = f_1(x_a) f_1(x_b) D_1(z_h), \qquad \boldsymbol{\nu} = (-1, -1, -2), \tag{30}$$

and the extracted hard coefficient was exactly independent of x_a, x_b, z_h. The NNLO card currently derives the valuation from the exact contribution sum; it should be made explicit only after the complete 342-master data establish a common value. The NNLO timing is a test of one two-contribution stored target, not an estimate for all 342 masters.

An independent end-to-end NLO calculation generated 16 cut integrals, reduced them to one master, and reconstructed the final expression exactly. Kira took 17.50 s and the coefficient stage took 10.34 s; the compact result occupied 122.48 KiB. This test exercises the same public simplification routine used for the stored-data calculations.

10 Eight-kernel time estimate

The selected post-IBP source occupies 859,107,253 serialized bytes. The old all-master post-IBP source occupies 3,960,695,102 bytes, giving the measured size ratio

$$\rho = \frac{3{,}960{,}695{,}102}{859{,}107{,}253} = 4.61. \tag{31}$$

Scaling the selected-column measurements by ρ gives

$$t_{\text{roots}} \simeq 4.61 \times 1{,}304.64\,\text{s} = 1.67 \text{ h}, \tag{32}$$

$$t_{\text{assemble+scale}} \simeq 4.61 \times (1{,}260.22 + 430.91 + 904.57)\,\text{s} = 3.32 \text{ h}. \tag{33}$$

Uneven column sizes, serialization, scheduling, and exact audits motivate a planning interval of

$$\boxed{\text{6–10 hours}} \tag{34}$$

for the complete post-IBP coefficient stage on eight kernels, starting from the existing Kira reduction. This extrapolation assumes that the selected columns are representative per serialized source byte; it has not yet been measured on all 342 columns.

A contemporaneous note quoted 1,313.82 s for one Kira calculation, but the matching timing log was not found during this audit. That number is not used in the estimate above.

There is no defensible time estimate for the *complete exact NNLO hard function*. Analytic evaluation of the 342 cut masters, boundary data, endpoint expansion, plus distributions, and combination with the remaining NNLO sectors are outside this one-day timing sample. A numerical master check cannot replace those analytic steps.

11 Unified simplification route

The physics sequence is common to NLO and NNLO:

1. read the complete kinematic, fraction, distribution, scale, forbidden variable, and branch declarations from the card;
2. substitute the exact hadronic coordinates;
3. derive or verify the common twist-two distribution product;
4. lift only certified half-integer Laurent monomials in positive fractions and prove Eq. (13);
5. derive or verify the common fraction valuation and reject any remaining forbidden variable;
6. derive each scale power from exact homogeneity, transform to the card's dimensionless invariants, and verify exact reconstruction.

Both input forms are converted to contribution groups before step 2 and then enter the same algebraic kernel. Consequently perturbative order does not select a different mathematical normal form. The only scheduling difference is whether contributions are summed before entering the routine or retained as a bounded list until their distribution and Laurent factors have been certified. When complete C_m expressions remain small, the assembled input in Eq. (28) is shorter and allows Mathematica to simplify each coefficient directly. When assembly is the memory bottleneck, the contribution-wise input performs bounded exact denominator algebra before the same physical checks. This scheduling decision should be made from measured expression size, not from the labels NLO and NNLO.

A Complete measured method inventory

The following tables include every method for which a numeric result was retained. A dash means that the corresponding metric was not measured or no result was produced.

Table 3: NLO UU simplification measurements.

Method	Object and metric	Input	Output	Time	Interpretation
Pair first	1332 objects, `ByteCount`	89,111,000	2,694,944	43.403 s	Repeats algebra; final master 1,574,064 bytes.
Target first	117 objects, `ByteCount`	46,940,920	1,362,792	23.715 s	Fastest first nontrivial route; final master 1,548,952 bytes.
Master first	one coefficient	–	–	> 281 s	Unfinished.
Target then master	8 objects, `ByteCount`	1,646,272	1,036,008	62.125 s	Second stage only; sequential total 85.840 s.
Largest target, termwise `Simplify`	`ByteCount`	3,517,376	49,016	4.838 s	Equality check later reached its time bound.
Largest target, whole `Simplify`	`ByteCount`	3,517,376	327,568	86.378 s	Slower and larger than termwise.
Largest target, whole `FullSimplify`	`ByteCount`	3,517,376	–	120.045 s	Unfinished.
Largest target, structural rational form	`ByteCount`	3,517,376	6,812,872	50.350 s	Expanded.
Common-factor extraction	`ByteCount`	1,362,792	684,744	3.393 s	Fraction-free targets.
Factored-route final cleanup	`ByteCount`	855,104	759,840	10.741 s	Full factor-first route about 37.76 s.
Second master sweep	`ByteCount`	759,840	759,840	9.426 s	No change.
Direct target cleanup	`ByteCount`	1,542,536	727,376	450.188 s	Much slower than factor-first.
33-rule certified normalization	`ByteCount`	1,035,984	283,144	9.437 s	Exact fraction independence.
Divide only	`ByteCount`	1,035,984	1,089,000	0.800 s	Fractions remain.
Divide then `Simplify`	`ByteCount`	1,035,984	283,144	5.877 s	Best compact strip variant.
Divide then `FactorTerms`	`ByteCount`	1,035,984	1,090,008	5.384 s	Fractions remain.
Final `FullSimplify`	`ByteCount`	283,144	256,456	32.352 s	Modest extra reduction.
Final `Factor`	`ByteCount`	283,144	925,968	0.146 s	Expanded.
Final color collection	`ByteCount`	283,144	562,496	0.185 s	Expanded.
Signature cleanup	`ByteCount`	1,035,984	3,879,816	1.868 s	Expanded 3.745-fold.
Strip only	`ByteCount`	1,035,984	1,935,248	1.741 s	Expanded.
Term cleanup	`ByteCount`	1,035,984	1,935,248	1.691 s	Expanded.
Denominator groups	`ByteCount`	1,035,984	1,937,768	1.905 s	Expanded.
Color collection variant	`ByteCount`	1,035,984	1,935,248	1.821 s	Expanded.
Automatic common factor	Syntactic factor	–	incomplete	0.0067 s	Missed $1/(x_a x_b)$.

Table 4: NLO TT and NNLO simplification measurements.

Method	Object and metric	Input	Output	Time	Interpretation
TT incorrect tensor route	`ByteCount`	–	184,796,536	–	Wrong dimension replacement; 39,036 GLIs.
TT corrected sparse cut route	`ByteCount`	–	27,754,864	–	Exact tensor tests; nine GLIs.
TT target first, 5 by 5	`ByteCount`	139,298,664	10,562,384	52.02 s	Four kernels.
TT final master cleanup	serialized bytes	–	750,504	30.36 s	Six masters.
TT angular master	`ByteCount`	1,224,256	119,520	1.874 s	Two angular coefficients.
64 largest NNLO targets	`ByteCount`	83,386,848	28,279,424	75.643 s	Four kernels; 2.949-fold.
Largest target, whole `Simplify`	`ByteCount`	2,434,656	811,576	15.30 s	Exact zero difference.
Largest target, termwise `Simplify`	`ByteCount`	2,434,656	701,328	6.92 s	Exact zero difference.
Largest target, `Cancel[Together]`	`ByteCount`	2,434,656	3,748,912	0.37 s	Expanded.
Largest target, add `FactorTerms`	`ByteCount`	2,434,656	5,316,312	0.83 s	Expanded further.

Method	Object and metric	Input	Output	Time	Interpretation
Whole-master `Simplify`	serialized bytes	4.7–129.8 MB	–	1,800 s each	Eight jobs unfinished; about 34 GiB total.
Master 8 termwise	serialized bytes	4,655,210	3,082,976	175.070 s	377 terms, eight kernels.
Master 1 signatures	serialized bytes	1,607,146,927	920,945,177	about 19.5 min	Rational normalization within signatures.
Master 2 signatures	serialized bytes	903,734,615	464,281,765	about 15 min	Rational normalization within signatures.
Master 4 signatures	serialized bytes	83,696,010	47,987,161	68.55 s	2,591 different signatures.
Old all-master cleanup	serialized bytes	3,960,695,102	3,597,827,125	–	Only 1.101-fold.
Symbolica then Mathematica	serialized MB	–	0.81 MB	12.16 s	Direct Mathematica: 0.70 MB in 5.91 s.
125-target root sample	`ByteCount`	52,630,888	559,984	30.45 s	Four kernels; certified root grammar.
Entrywise root cleanup	serialized bytes	859,107,253	131,787,741	1,304.639 s	Three post-IBP master columns, eight kernels.
Eager denominator merge	`ByteCount`	422,282,520	1,727,567,608	6,375.22 s	6,290 fractions; expanded strongly.
Size-monotone merge	`ByteCount`	422,282,520	351,559,840	1,260.216 s	6,557 fractions; serialized 99,892,985 bytes.
Dimensionless scale separation	`ByteCount`	351,559,840	296,802,344	430.914 s	Serialized 99,892,985 to 92,385,524.
Dimensionless denominator merge	`ByteCount`	296,802,344	296,275,768	904.575 s	5,973 fractions; serialized 92,241,826.
Denominator-factor inventory	serialized bytes	92,241,826	91,385,453	27.94 s	Only 0.93 percent smaller; 30 factors.

B Provenance and retained records

The documentation directory contains:

- the two earlier narrative records;
- the compact NLO UU, TT, and NNLO manifests quoted above;
- scripts for every principal benchmark and branch-safe transformation;
- a machine-readable CSV table naming every size metric;
- an artifact index pointing to the original large expressions and their stored hashes.

The report does not duplicate multi-gigabyte coefficient files.